

A low-fat diet and/or strenuous exercise alters the IGF axis in vivo and reduces prostate tumor cell growth in vitro.

.

BACKGROUND: Prostate cancer is the most common solid-tumor cancer in US males but is rare in Asian males. When Asian men adopt the US lifestyle, clinical prostate cancer increases greatly. Epidemiological data from men in the US indicate that regular activity may reduce the risk for prostate cancer. METHODS: Serum was obtained from three groups of similar-aged men, Control, Diet and Exercise, and Exercise alone were used to stimulate LNCaP cells in culture. Growth and apoptosis of tumor cells were measured. Serum samples were also used to measure insulin, IGF-1, IGFBP-1. RESULTS: The Diet and Exercise and the Exercise alone groups had lower serum insulin and IGF-1 but higher IGFBP-1 compared to Controls. LNCaP cell growth was reduced in both groups compared to Control and there was a major increase in apoptosis of tumor cells. CONCLUSIONS: A low-fat diet and/or intensive exercise results in change in serum hormones and growth factors in vivo that can reduce growth and induce apoptosis of LNCaP prostate tumor cells in vitro. Copyright 2003 Wiley-Liss, Inc.